3. Develop a flowchart in Figma depicting the operation of a Network Switching Subsystem in a mobile cellular system.

Aim:

To develop a flowchart in Figma representing the operation of the Network Switching Subsystem in a mobile cellular system, including subscriber identification, call routing, connection establishment, and call termination.

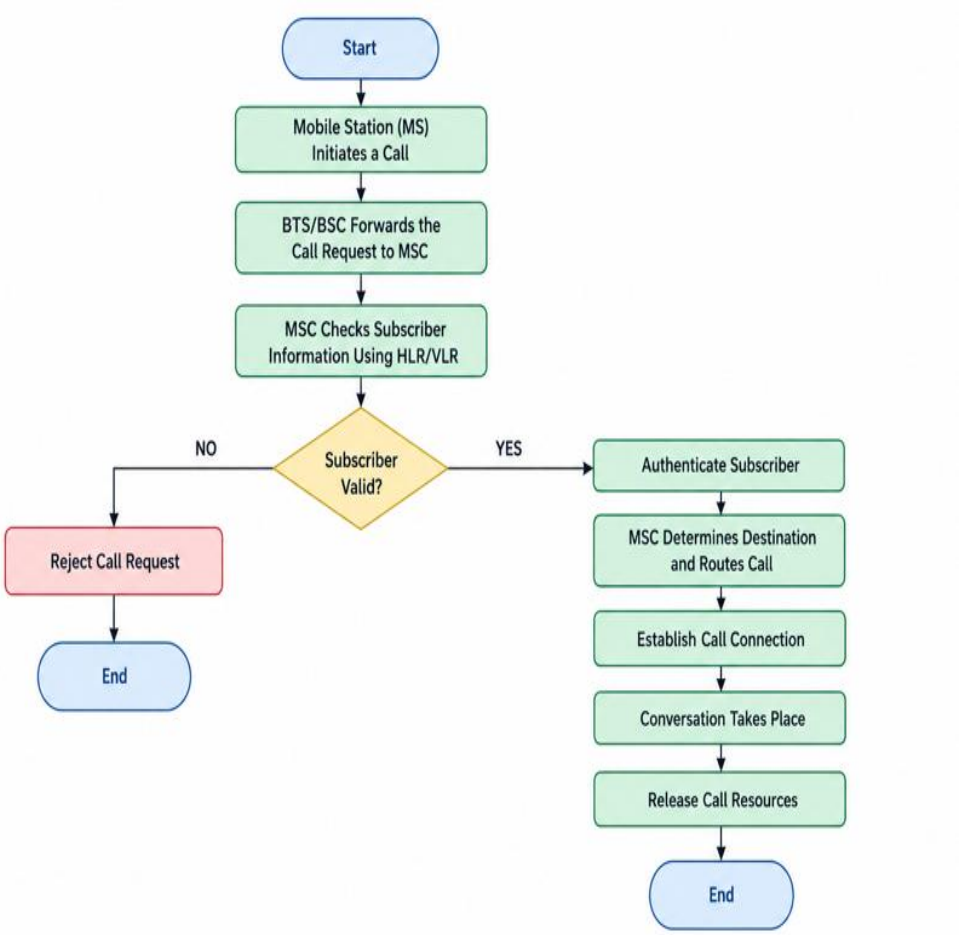
Software Used:

- Figma — to create the flowchart using shapes, connectors, and text.
- Web browser — to access Figma.

Algorithm

- Start.
- The mobile station initiates a call or data request.
- The request passes through the Base Transceiver Station (BTS) and Base Station Controller (BSC).
- The request reaches the NSS through the Mobile Switching Centre (MSC).
- The NSS identifies whether the request is a voice call or a data session.
- For a voice call, check subscriber details using the VLR and HLR, authenticate the user, and establish the call.
- For a data session, activate the data context and route data through the appropriate packet network
- The NSS performs switching and routing to connect the required destination.
- Establish the requested service.
- End.

Flowchart Design Output:



Result:

Thus, a flowchart illustrating the operation of the Network Switching Subsystem in a mobile cellular system was designed. It demonstrates subscriber verification, call processing, switching, routing, and data-session establishment using the major network components.